

CADORA

Problem Statement:

The luxury market is currently filled with high-fidelity fakes and the resale of stolen goods, both of which erode brand equity and consumer trust. Because a product's value is currently tied solely to its **physical possession**, any individual holding the item (irrespective of legitimacy in ownership) effectively claims its full market value.

The problem lies in the thought that possession equals ownership.

Existing Solutions:

Currently, the luxury industry relies on **laser-etched serial numbers** and physical **paper-based certificates** to serve as the primary proof of a product's legitimacy.

In many cases, these serial numbers are cross-referenced with **brand databases** where authorized dealers register the initial sale to track warranty and service history.

The most advanced modern standard involves **NFC-enabled warranty cards** used by brands like Rolex which allow owners to scan the card with a smartphone to establish a secure, digital link directly to the manufacturer's official records. While these methods effectively verify that an item's genuinity at the point of manufacture, they do not autonomously track subsequent changes in ownership, and it also doesn't say about who owns it.

Inefficiency of current solutions :

The core vulnerability of existing solutions is that any physical evidence, no matter how sophisticated it is, can be compromised by a con artist with technical knowhow. High-security paper certificates, holographic seals, and laser-etched serial numbers are static objects. Once a con artist masters the replication process of these physical markers, they don't just fake the product; they fake the **proof of legitimacy** itself.

This creates a question of, "Can you even trust the proof of legitimacy? Or is the proof itself fake?".

Furthermore, it isn't just the proof of legitimacy that matters, proof of ownership too must be taken into consideration.

Current NFC-based solutions, such as those used by Rolex, provide only a "Partial Truth." While an NFC scan may verify that a watch was manufactured by the brand, it remains silent on the most critical question: **"Who is the current rightful owner?"** Which means, anyone who steals the Rolex can decide to sell it as if they owned it.

This is because:

- The best systems typically only record the initial sale from the authorized dealer to the first owner.
- **Successor issue:** As the asset moves through the secondary market from the initial owner to successive owners, the "track of truth" is lost.
- **The Thief's Advantage:** If a watch is stolen from the successor, the thief can still present the "Authentic" NFC tag to a buyer. Since the system doesn't track current ownership, the tag actually helps the thief by proving the item is real, without ever revealing it is stolen.

Our Solution:

Our solution introduces a **Digital-product-pass** strategy powered by a brand managed private blockchain. Under this system, a physical luxury asset is "half a key." The second half exists as a non-fungible digital asset (NFT) on the brand's secure, immutable network.

Every luxury item is issued with a unique NFT at the point of manufacture. This digital asset serves as a comprehensive "Living Record," containing:

- **Core Metadata:** Product type, specific make, and a serial number that is physically and cryptographically linked to the item.
- **The Chain of ownership:** A verified history of ownership.
- **Dynamic Appendage:** Every subsequent change in ownership is appended to this record only after a verified transaction occurs, and that too can only be changed by the current owner by contacting the brand itself to change / add an owner.

Ownership management:

Ownership is not transferred simply by handing over the physical item. To update the "Chain of Title," the current owner must interact with the brand's infrastructure to request an update. This ensures that the digital title only moves when the rightful owner authorizes it, making it impossible to break or forge.

Instant verification:

The system enables "On-the-Spot" verification. A simple scan of a secure QR code or embedded chip on the product immediately queries the brand's ledger. If the person possessing the item does not match the owner recorded on the blockchain, the item is instantly flagged as illegitimate or unauthorized.

Conclusion:

By removing the ability to prove ownership without the digital record, we kill the profit motive for luxury crime. If a stolen item can't be resold and a fake can't be verified, the criminal market collapses. This creates a self-protecting system where financial loss is prevented because authenticity is no longer a guess, it is a certificate that cannot be tampered.